

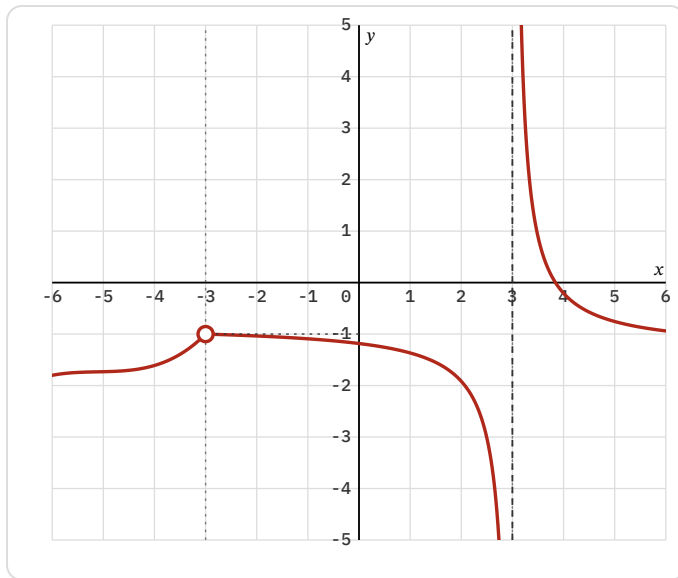
Mock Midterm A — Answer Key

Practice midterm (AI-Free) · Sections 1.1–1.4

Total: 91 points

1.

Use the graph of $y = f(x)$ to state the value of each quantity, if it exists. If it does not exist, write **DNE**.



a.) $f(-3)$ (1)

DNE

b.) $\lim_{x \rightarrow -3^-} f(x)$ (1)

-1

c.) $\lim_{x \rightarrow -3^+} f(x)$ (1)

-1

d.) $\lim_{x \rightarrow -3} f(x)$ (1)

-1

e.) $f(3)$ (1)

DNE

f.) $\lim_{x \rightarrow 3^-} f(x)$ (1)

$-\infty$

g.) $\lim_{x \rightarrow 3^+} f(x)$ (1)

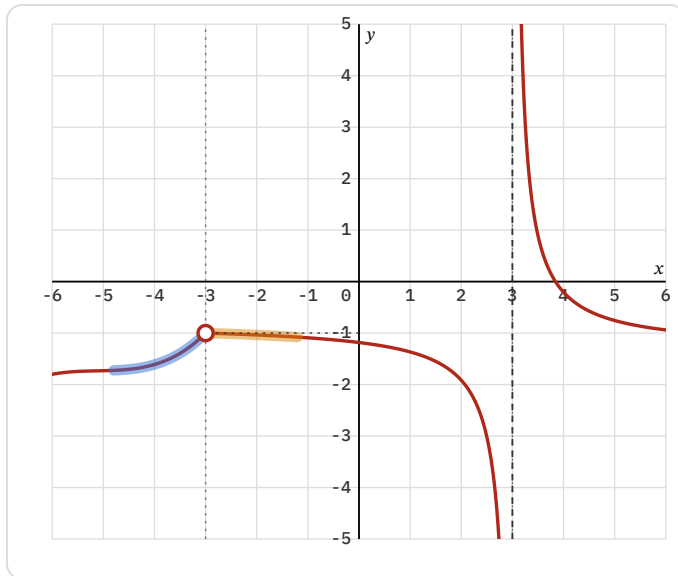
∞

h.) $\lim_{x \rightarrow 3} f(x)$ (1)

DNE

8 pts

WORKED SOLUTION



— approaching $x = -3$ from the left — from the right. Follow each colored piece of the curve to the height it heads toward.

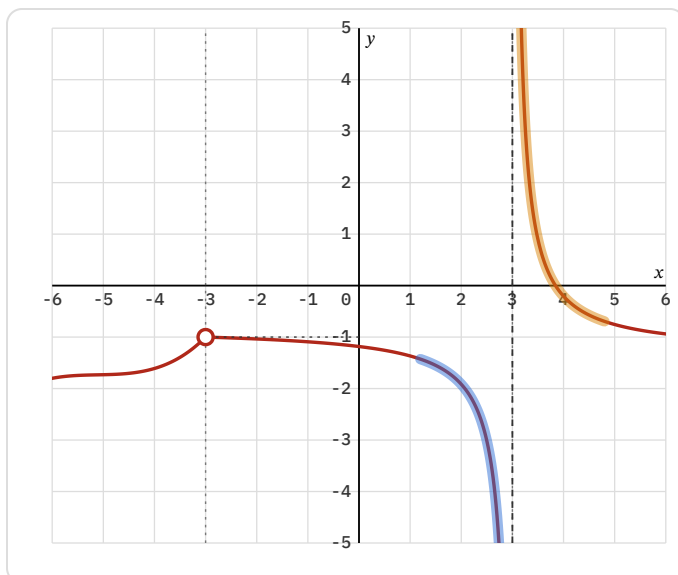
At $x = -3$

There is no filled dot on the line $x = -3$, so $f(-3)$ is undefined $\rightarrow$ DNE.

Tracing the curve toward $x = -3$ from the left, the heights approach -1 (open or closed dot doesn't matter).

Tracing the curve toward $x = -3$ from the right, the heights approach -1 (open or closed dot doesn't matter).

Both one-sided limits equal -1, so $\lim_{x \rightarrow -3} f(x) = -1$ — even though $f(-3)$ is undefined. The limit ignores the value AT the point.



— approaching $x = 3$ from the left — from the right. Follow each colored piece of the curve to the height it heads toward.

At $x = 3$

There is no filled dot on the line $x = 3$, so $f(3)$ is undefined $\rightarrow$ DNE.

From the left, the curve runs along the asymptote $x = 3$ downward, so the limit is $-\infty$.

From the right, the curve runs along the asymptote $x = 3$ upward, so the limit is ∞ .

The left and right limits don't match, so $\lim_{x \rightarrow 3} f(x)$ DNE.

2. Let $f(x) = \frac{x-4}{\sqrt{x}-2}$. Use the table to see what the outputs do as x approaches 4. 7 pts

x	3.5	3.9	3.99	4	4.01	4.1	4.5
$f(x)$	3.870829	3.974842	3.997498	?	4.002498	4.024846	4.12132

- a.) Can you find $\lim_{x \rightarrow 4} f(x)$ just by plugging in $x = 4$? (1)

(B) No — $x = 4$ makes a denominator 0, so $f(4)$ is undefined

- b.) As $x \rightarrow 4^-$, $f(x)$ approaches... (2)

4

- c.) As $x \rightarrow 4^+$, $f(x)$ approaches... (2)

4

- d.) $\lim_{x \rightarrow 4} f(x) =$ (2)

4

WORKED SOLUTION

Reading the left three columns, the outputs head toward 4; the right three head toward 4.

Both sides agree, so the limit is 4.

$$\lim_{x \rightarrow 4} \frac{x-4}{\sqrt{x}-2} = 4$$

Why: $\frac{(\sqrt{x}-2)(\sqrt{x}+2)}{\sqrt{x}-2} = \sqrt{x} + 2 \rightarrow 4$

3.

Suppose

4 pts

$$\lim_{x \rightarrow 5} f(x) = 3, \quad \lim_{x \rightarrow 5} g(x) = 5, \quad \lim_{x \rightarrow 5} h(x) = 16$$

Use the properties of limits to find each limit.

a.) $\lim_{x \rightarrow 5} x^2 f(x)$ (1)

75

b.) $\lim_{x \rightarrow 5} \sqrt{h(x)} f(x)$ (1)

12

c.) $\lim_{x \rightarrow 5} ([f(x)]^3 - g(x))$ (1)

22

d.) $\lim_{x \rightarrow 5} [2f(x) + g(x)]^2$ (1)

121

WORKED SOLUTION

a. $\lim_{x \rightarrow 5} x^2 f(x) = (5)^2 \cdot (3) = 75$

b. $\lim_{x \rightarrow 5} \sqrt{h(x)} f(x) = \sqrt{16} \cdot (3) = 4 \cdot (3) = 12$

c. $\lim_{x \rightarrow 5} ([f(x)]^3 - g(x)) = (3)^3 - (5) = 27 - 5 = 22$

d. $\lim_{x \rightarrow 5} [2f(x) + g(x)]^2 = [2(3) + (5)]^2 = (11)^2 = 121$

The quotient property works here because the bottom limit isn't 0.

4.

Calculate each limit algebraically. (Direct substitution gives $\frac{0}{0}$ — simplify first.) If it doesn't exist, write DNE.

a.) $\lim_{h \rightarrow 0} \frac{\frac{1}{3+h} - \frac{1}{3}}{h}$ (2)
 $-\frac{1}{9}$

b.) $\lim_{x \rightarrow -2} \frac{3x^2 + 3x - 6}{2x^2 + 7x + 6}$ (2)
 9

4 pts

WORKED SOLUTION

a.

$$\lim_{h \rightarrow 0} \frac{\frac{1}{3+h} - \frac{1}{3}}{h} = \lim_{h \rightarrow 0} \frac{\frac{3-(3+h)}{3(3+h)}}{h} = \lim_{h \rightarrow 0} \frac{-h}{3(3+h)h} = \lim_{h \rightarrow 0} \frac{-1}{3(3+h)} = -\frac{1}{9}$$

b.

$$\lim_{x \rightarrow -2} \frac{3x^2 + 3x - 6}{2x^2 + 7x + 6} = \lim_{x \rightarrow -2} \frac{(x+2)(3x-3)}{(x+2)(2x+3)} = \lim_{x \rightarrow -2} \frac{3x-3}{2x+3} = \frac{-9}{-1} = 9$$

Notation tip: keep writing **lim** on every line until the step where you actually substitute.

5. Compute each limit algebraically. Use ∞ / $-\infty$ for unbounded results.

6 pts

a.) $\lim_{x \rightarrow -\infty} \frac{4x^2}{-x^4 - 7x^2 + 4x}$ (2)
0

b.) $\lim_{x \rightarrow \infty} \frac{5x}{4x - 5}$ (2)
 $\frac{5}{4}$

c.) $\lim_{x \rightarrow \infty} \frac{4x^3 - 6x^2}{-4x^2}$ (2)
 $-\infty$

WORKED SOLUTION

a. Divide every term by x^4 , the highest power of x in the denominator:

$$\lim_{x \rightarrow -\infty} \frac{4x^2}{-x^4 - 7x^2 + 4x} = \lim_{x \rightarrow -\infty} \frac{\frac{4}{x^2}}{-1 - \frac{7}{x^2} + \frac{4}{x^3}}$$

The top $\rightarrow 0$ while the bottom $\rightarrow -1$, so the limit is 0.

b. Divide every term by x , the highest power of x in the denominator:

$$\lim_{x \rightarrow \infty} \frac{5x}{4x - 5} = \lim_{x \rightarrow \infty} \frac{5}{4 - \frac{5}{x}}$$

Every fraction term $\rightarrow 0$, leaving $\frac{5}{4} = \frac{5}{4}$.

c. Divide every term by x^2 , the highest power of x in the denominator:

$$\lim_{x \rightarrow \infty} \frac{4x^3 - 6x^2}{-4x^2} = \lim_{x \rightarrow \infty} \frac{4x - 6}{-4}$$

The bottom $\rightarrow -4$, but the top is dominated by $4x^1$, which $\rightarrow \infty$ as $x \rightarrow \infty$; dividing by -4 gives $-\infty$.

6.

Let

7 pts

$$G(x) = \begin{cases} x^2 + 2 & x < 0 \\ 2x + 2 & x > 0 \end{cases}$$

Find each quantity. If it does not exist, write DNE.

a.) $G(0)$ (1)

DNE

b.) $\lim_{x \rightarrow 0^-} G(x)$ (1)

2

c.) $\lim_{x \rightarrow 0^+} G(x)$ (1)

2

d.) $\lim_{x \rightarrow 0} G(x)$ (1)

2

e.) $G(2)$ (1)

6

f.) $\lim_{x \rightarrow -\infty} G(x)$ (1)

 ∞

g.) $\lim_{x \rightarrow \infty} G(x)$ (1)

 ∞ **WORKED SOLUTION** **$G(0)$:** the pieces cover $x < 0$ and $x > 0$ only, so $G(0)$ is undefined $\rightarrow$ DNE.**Left:** use $x^2 + 2$: $(0)^2 + 2 = 2$.**Right:** use $2x + 2$: $2(0) + 2 = 2$.**Two-sided:** both sides give 2, so the limit is 2 (even though $G(0)$ is undefined). **$G(2)$:** $2 > 0$, so use $2x + 2$: $2(2) + 2 = 6$.**End behavior:** as $x \rightarrow -\infty$ we're on the $x^2 + 2$ piece — an even-degree term with positive leading coefficient $\rightarrow \infty$. As $x \rightarrow \infty$ we're on the line with slope 2 $\rightarrow \infty$.

7. Let

6 pts

$$f(x) = \begin{cases} \frac{2x^2 + 11x + 15}{2x + 5} & x < -\frac{5}{2} \\ 3x + 8 & x > -\frac{5}{2} \end{cases}$$

Find each limit or function value algebraically. Write DNE if it doesn't exist.

a.) $\lim_{x \rightarrow 1} f(x)$ (1)

11

b.) $\lim_{x \rightarrow -5} f(x)$ (1)

-2

c.) $f\left(-\frac{5}{2}\right)$ (1)

DNE

d.) $\lim_{x \rightarrow -\frac{5}{2}^-} f(x)$ (1)

$\frac{1}{2}$

e.) $\lim_{x \rightarrow -\frac{5}{2}^+} f(x)$ (1)

$\frac{1}{2}$

f.) $\lim_{x \rightarrow -\frac{5}{2}} f(x)$ (1)

$\frac{1}{2}$

WORKED SOLUTION

For $x < -\frac{5}{2}$, factor the top: $\frac{(2x+5)(x+3)}{2x+5} = x + 3$ (valid since $x \neq -\frac{5}{2}$ there).

a. $1 > -\frac{5}{2} \rightarrow$ bottom piece: $3(1) + 8 = 11$.

b. $-5 < -\frac{5}{2} \rightarrow$ top piece, simplified: $-5 + 3 = -2$.

c. Neither piece includes $x = -\frac{5}{2}$, so $f\left(-\frac{5}{2}\right)$ is undefined $\rightarrow$ DNE.

d. Left: $\lim_{x \rightarrow -\frac{5}{2}^-} (x + 3) = \frac{1}{2}$.

e. Right: $\lim_{x \rightarrow -\frac{5}{2}^+} (3x + 8) = \frac{1}{2}$.

f. Both sides equal $\frac{1}{2}$, so the limit is $\frac{1}{2}$.

8.

Let

5 pts

$$f(x) = \begin{cases} -x^2 + 4x & x \leq 0 \\ -3x & x > 0 \end{cases}$$

Determine whether $f(x)$ is continuous at $x = 0$.

a.) $f(0)$ (1)

0

b.) $\lim_{x \rightarrow 0^-} f(x)$ (1)

0

c.) $\lim_{x \rightarrow 0^+} f(x)$ (1)

0

d.) $\lim_{x \rightarrow 0} f(x)$ (1)

0

e.) Is f continuous at $x = 0$? (1)

(A) Yes — $f(0)$ exists, the limit exists, and they are equal

WORKED SOLUTION

1. $f(0)$: 0 satisfies $x \leq 0$, so use the top piece: $-(0)^2 + 4(0) = 0$.

2. **Limit**: left uses the top piece $\rightarrow 0$; right uses $-3x \rightarrow -3(0) + 0 = 0$.

3. Both one-sided limits are 0, so the limit exists and equals $f(0)$. **Continuous**.

9.

Classify each function at the given x -value.

3 pts

a.) $f(x) = \begin{cases} x - 2 & x < -1 \\ x - 5 & x \geq -1 \end{cases}$ at $x = -1$ (1)

(C) Jump discontinuity

b.) $f(x) = 3x^2 + x + 4$ at $x = 4$ (1)

(A) Continuous

c.) $f(x) = \frac{x+2}{x^2-4}$ at $x = 2$ (1)

(D) Infinite discontinuity

WORKED SOLUTION

a. Left $\rightarrow -3$, right $\rightarrow -6$. Different finite values $\rightarrow$ jump.

b. Polynomials are continuous everywhere.

c. Factor: $\frac{(x+2)}{(x+2)(x-2)}$. The $(x - 2)$ factor does NOT cancel, so $x = 2$ is a vertical asymptote $\rightarrow$ infinite.

10.

Let $s(t) = -16t^2 + 96t + 34$, where $s(t)$ represents the height, in feet, of a ball t seconds after it is thrown upward. Find the average rate of change over $[1, 3]$.

a.) $s(1)$ (2)

114

b.) $s(3)$ (2)

178

c.) Average rate of change (number only) (2)

32

d.) Which sentence correctly states the result **with units**? (1)

(B) Between $t = 1$ and $t = 3$ seconds, the height of the ball changes at an average rate of 32 feet per second.

7 pts

WORKED SOLUTION

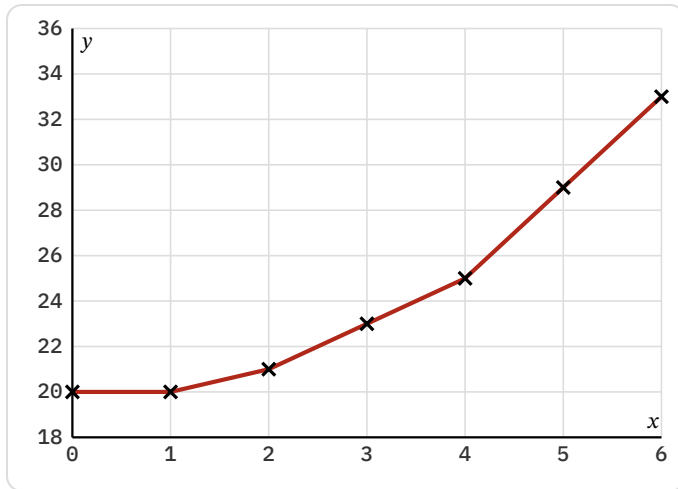
$$s(1) = 114, \quad s(3) = 178$$

$$\frac{s(3) - s(1)}{3 - 1} = \frac{178 - (114)}{2} = \frac{64}{2} = 32 \text{ feet per second}$$

It's an *average* over an interval (not an instant), and the units are feet per second — feet per second. The total change would be 64 feet.

11.

The graph shows the closing price (in dollars) of a stock at the start of each month in 2026, where $x = 0$ is January, $x = 1$ is February, and so on.



a.) Average rate of change on $[0, 4]$ (dollars per month) (2)

$$\frac{5}{4}$$

b.) On which interval is the average rate of change greatest? (1)

(C) $[4, 6]$

3 pts

WORKED SOLUTION

$$\frac{f(4) - f(0)}{4 - 0} = \frac{25 - 20}{4} = \frac{5}{4} \text{ dollars per month}$$

Between Jan and May the price changed by $\frac{5}{4}$ dollars per month on average — the slope of the secant line joining those two points.

Greatest-rate check: $[0, 2]: \frac{1}{2}$, $[2, 4]: 2$, $[4, 6]: 4 \rightarrow [4, 6]$.

12. Let $f(x) = x^3$. Find and simplify the difference quotient $\frac{f(x+h)-f(x)}{h}$, then use it to fill in the table.

a.) Simplified difference quotient (in terms of x and h) (3)

$$3x^2 + 3xh + h^2$$

b.) $x = -1, h = 1$ (1)

$$1$$

c.) $x = -1, h = 0.1$ (1)

$$\frac{271}{100}$$

d.) $x = -1, h = 0.01$ (1)

$$\frac{29701}{10000}$$

e.) As $h \rightarrow 0$, what value do these approach? (1)

$$3$$

7 pts

WORKED SOLUTION

$$f(x+h) = (x+h)^3 = x^3 + 3x^2h + 3xh^2 + h^3$$

$$\frac{f(x+h) - f(x)}{h} = \frac{3x^2h + 3xh^2 + h^3}{h} = 3x^2 + 3xh + h^2$$

Now substitute:

$$x = -1, h = 1 : 1$$

$$x = -1, h = 0.1 : \frac{271}{100} = 2.71$$

$$x = -1, h = 0.01 : \frac{29701}{10000} = 2.9701$$

Letting $h \rightarrow 0$ leaves **3**: the instantaneous rate of change (slope of the tangent line) at $x = -1$. That's exactly what section 1.4 formalizes.

13.

Let $f(x) = -x^2 + x$. Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$.

a.) $f'(x) =$ (3)
 $-2x + 1$

b.) $f'(-2) =$ (2)
 5

c.) Slope of the tangent line to f at $x = 2$ (2)
 -3

d.) What does $f'(-2) = 5$ mean? (1)

(B) The instantaneous rate of change of f at $x = -2$ is 5 — the slope of the tangent line there.

8 pts

WORKED SOLUTION

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{[-(x+h)^2 + (x+h)] - (-x^2 + x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-x^2 - 2xh - h^2 + x + h + x^2 - x}{h} \\ &= \lim_{h \rightarrow 0} \frac{-2xh - h^2 + h}{h} = \lim_{h \rightarrow 0} \frac{h(-2x - h + 1)}{h} \\ &= \lim_{h \rightarrow 0} (-2x - h + 1) = -2x + 1 \end{aligned}$$

Then $f'(-2) = 5$ and the slope of the tangent at $x = 2$ is $f'(2) = -3$.

$f'(-2)$ is the *instantaneous* rate of change at one input — the slope of the tangent line, not the height of the graph and not an average over an interval.

14.

Let $f(x) = -2x^2 + 5x + 5$, so $f'(x) = -4x + 5$.

6 pts

a.) At what x is the tangent line horizontal? (2)

$$\frac{5}{4}$$

b.) What is the y -coordinate of that point? (2)

$$\frac{65}{8}$$

c.) At what x does the tangent line have slope -2 ? (2)

$$\frac{7}{4}$$

WORKED SOLUTION

$$f'(x) = 0 : \quad -4x + 5 = 0 \Rightarrow x = \frac{5}{4}$$

$$f\left(\frac{5}{4}\right) = \frac{65}{8}$$

$$f'(x) = -2 : \quad -4x + 5 = -2 \Rightarrow x = \frac{-7}{-4} = \frac{7}{4}$$

15. Let $f(x) = x^2 - 3$. Complete the limit definition, then interpret $f'(-3)$. 6 pts

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{((x+h)^2 - 3) - (x^2 - 3)}{h} = ?$$

a.) $f'(x) =$ (3)

$2x$

b.) $f'(-3)$ (2)

-6

c.) What does $f'(-3) = -6$ tell you? (1)

(A) The slope of the tangent line to the graph of f at $x = -3$ is -6.

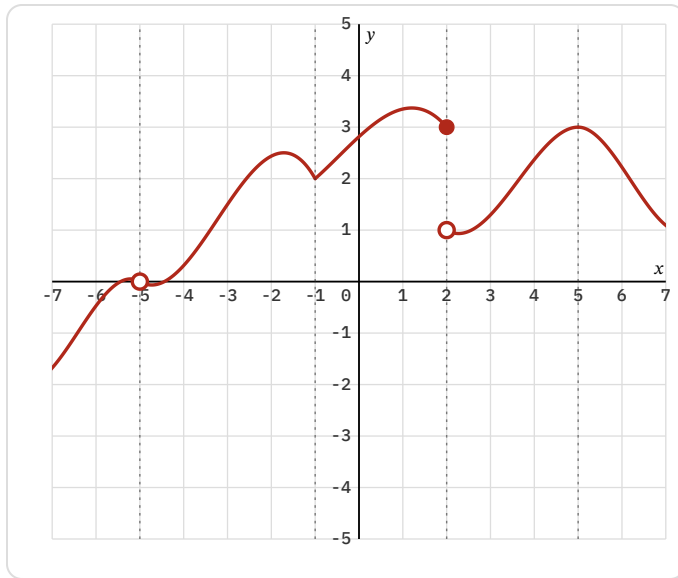
WORKED SOLUTION

$$\lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 3 - x^2 + 3}{h} = \lim_{h \rightarrow 0} \frac{h(2x + h)}{h} = \lim_{h \rightarrow 0} (2x + h) = 2x$$

$f'(-3) = 2(-3) = -6$. The graph of f has an instantaneous rate of change of -6 at $x = -3$: the tangent line there has slope -6 (f is decreasing).

16.

At each labeled x -value, decide whether the graph of f is continuous there and whether it is differentiable there.



- a.) At $x = -5$ (1)
(C) Not continuous (so not differentiable)
- b.) At $x = -1$ (1)
(B) Continuous but NOT differentiable
- c.) At $x = 2$ (1)
(C) Not continuous (so not differentiable)
- d.) At $x = 5$ (1)
(A) Continuous and differentiable

4 pts

WORKED SOLUTION

$x = -5$: There's a hole (the dot isn't on the curve) $\rightarrow$ not continuous, so not differentiable.

$x = -1$: No break, but the curve comes to a sharp point (corner): the slopes from the left and right disagree $\rightarrow$ continuous, not differentiable.

$x = 2$: The graph jumps $\rightarrow$ not continuous, so not differentiable.

$x = 5$: Smooth and unbroken — a single tangent line fits $\rightarrow$ continuous and differentiable.
